



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science Engineering.
Summer Term Examination
Academic Year: 2024-25

Class: Third Year

Course: Combinatorics, Probability and Statistics. (Elective)

Course Code: PEC-CS317.

Date: 23-Jun-25

Time: 10:00 13:00

Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

- Q.1 A certain type of electric light bulb has a mean lifetime of 1500 hours and a standard deviation of 150 hours. Three bulbs are connected so that when one burns out, another will go on. Assuming the lifetimes are normally distributed, what is the probability that lighting will take place for (a) at least 5000 hours, (b) at most 4200 hours? 10
- Q.2 A sample of 10 measurements of the diameter of a sphere gave a mean inches and a standard deviation inch. Find (a) 95%, (b) 99% confidence limits for the actual diameter. 10
- Q.3 A random sample of size 100 is taken from an infinite population having mean $\mu=76$ and variance $\sigma^2=256$. What is the probability that mean μ will be between 75 and 78. 10
- Q.4 Explain Binomial distribution and Poissons distribution with appropriate formulas, expressions and examples. State and explain when they are useful at what circumstances. 10

Solve any Three from following. (20 x 3=60)

- Q.5 a) Explain the process of sampling and the different types of sampling. How large and small samples are differentiated? 07
- b) Explain point estimation and interval estimates. Explain how the confidence intervals for means are computed. Consider separately small and large samples ($n > 30$). 07
- c) A continuous random variable X has density function given by 06

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Find

(i) $E(X)$.

(ii) $E(X^2)$.

(iii) variance of X and

(iv) standard deviation of X.

- Q.6 a) Explain point estimation and interval estimates. Explain how the confidence intervals for means are computed. Consider separately small and large samples ($n \geq 30$). 07
- b) Design a decision rule to test the hypothesis that a coin is fair if a sample of 64 tosses of the coin is taken and if a level of significance of (i) 0.05, (ii) 0.01 is used. 07
- c) Prove that $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ 06
- Q.7 a) A sample poll of 100 voters chosen at random from all voters in a given district indicated that 55% of them were in favor of a particular candidate. 07
- Find (i) 95%, (ii) 99% (iii) 99.73% confidence limits for the proportion of votes in favor of this candidate.
- b) Two samples of sizes 16 and 10, respectively, are drawn at random from two normal populations. If their variances are found to be 24 and 18, respectively, find (a) 98%, (b) 90% confidence limits for the ratio of the variances. 07
- c) Design a decision rule to test the hypothesis that a coin is fair if a sample of 64 tosses of the coin is taken and if a level of significance of (i) 0.05, (ii) 0.01 is used. 06
- Q.8 a) Assume that the heights of 3000 male students at a university are normally distributed with mean 68.0 inches and standard deviation 3.0 inches. If 80 samples consisting of 25 students each are obtained, what would be the mean and standard deviation of the resulting sample of means if sampling were done (i) with replacement, (ii) without replacement? 07
- b) An item is produced in large numbers. The machine is known to produce 5% defective. A quality control inspector is examining the items by taking at random. What is the probability that at least 4 items are to be examined in order to get two defectives? 07
- c) Suppose that the heights of 100 male students from XYZ University represent a random sample of the heights of all 1546 male students at university. Determine unbiased and efficient estimates of 06

- (i) the true mean
(ii) the true variance.

Heights(cms)	Class marks	Frequency(f)	f_x
155-159	157	5	785
160-164	162	18	2916
165-169	167	42	7014
170-174	172	27	4644
175-179	174	8	1416



**Shri Guru Gobind Singhji Institute of Engineering
and Technology, Nanded-431606**

Summer Term - Exam

Subject: Fundamentals of Artificial Neural Network

Code: **PEC-CS316**

Date: 23/06/2025

Time: 10.00 To 13.00

Max. Marks: 100

Instructions:

1. All questions are compulsory.
2. Assume suitable data if necessary.

Q.1 Write in brief to solve the Following (Any Four)

32

- a. Explain Feed Forward Neural network.
- b. Explain Gradient Descent rule and parameter updation rule.
- c. What does it mean to train the network? Explain it with example.
- d. Explain Taylor series.
- e. What are the limitations of single layer Perceptron?

Q.2 a. How to choose Loss function? Explain with example.

10

- b. What is a Sigmoid Neuron? Explain similarities & difference between sigmoid and Perceptron. 10
- c. What is a Multilayer Perceptron? Explain characteristics of Multilayer Perceptron. 10

Q.3 a. Explain how back propagation works with example.

10

- b. What is gradient descent? Explain the variants of Gradient Descent algorithm. 8
- c. Design NOR gate neural representation using MP model. 10
- d. Explain McCulloch-Pitts Model with example. 10

End



**SGGS Institute of Engineering and Technology,
Vishnupuri, Nanded**

Summer Term Examination - 2025

Third Year

Computer Science & Engineering

**Data Communication &
Networking**

PCC-CS311

**Date:
21/06/2025**

Time: 10:00 to 13:00

Max. Marks : 100

- Note: i. All questions are compulsory.
ii. Assume suitable data if necessary
iii. Use of non-programmable calculator is permitted.
iv. Figures to the right indicate Marks

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|--------|---|----|
| Q.1 a. | State and explain the components of data communication system. | 08 |
| b. | Explain Simplex, Half duplex and Full duplex communication. | 08 |
| c. | Explain calculation of data rate using Nyquist bit rate and Shannon capacity. Give Suitable examples. | 10 |
| Q.2 a. | State and explain the common characteristics of Line Coding. | 08 |
| b. | Explain NRZ-L and NRZ -I schemes of line coding. | 10 |
| c. | Explain Analog-to-Digital Conversion using PCM. | 08 |
| Q.3 a. | Explain BPSK and QPSK techniques. | 08 |
| b. | Define FHSS and Explain how it achieves spreading. | 08 |
| c. | How is the Even Parity System used for error detection? Give suitable example. | 08 |
| Q.4 a. | How is the Checksum used for error detection? Give suitable example. | 08 |
| b. | Explain the Polling method used for Media Access Control. | 08 |
| c. | Explain Ethernet frame structure and addressing. | 08 |
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Class: TY

Course: Compiler Design

Course Code: PCC-CS310

Date: 20/06/2025

Time: 10:00 to 01:00

Max. Marks: 100

Note: Assume suitable data whenever necessary and state it clearly.

Q.N.	Attempt any 10 questions from the following. Each carries 10 marks.	CO	BT
Q.1	Explain in detail the working of a compiler with a suitable example.	CO1	BT2
Q.2	Given the grammar: $S \rightarrow AaAb \mid BbBa$, $A \rightarrow \epsilon$, $B \rightarrow \epsilon$ i) Compute FIRST() and FOLLOW() functions ii) Construct a predictive parsing table iii) Parse the input string "ab".	CO3	BT3
Q.3	Compare top-down parsing and bottom-up parsing methods. What are LR parsers? Explain the LR parsing algorithm using a diagram.	CO2	BT2
Q.4	What is code generation? Explain it with a suitable example and algorithm.	CO2	BT1
Q.5	Consider the following grammar for a simple arithmetic language. $S \rightarrow E$ $E \rightarrow E + T \mid E - T \mid T$ $T \rightarrow T * F \mid T / F \mid F$ $F \rightarrow (E) \mid id \mid num$ Design a syntax-directed translation scheme to generate intermediate code for the arithmetic expression $id + id * num$ and show the parse tree, semantic rules, and intermediate code generated.	CO3	BT6
Q.6	Explain different simplification methods for context-free grammars with suitable examples.	CO5	BT2
Q.7	What is an activation record? What do its fields represent? Explain in detail.	CO4	BT2
Q.8	What is meant by handle pruning in bottom-up parsing? Explain with the help of an example. Also, construct a bottom-up parse tree for the input string $w=aaaa++$ using the grammar: $S \rightarrow SS + \mid SS * \mid a$	CO3	BT3
Q.9	Let G be the grammar: $S \rightarrow aB \mid bA$ $A \rightarrow a \mid aS \mid bAA$ $B \rightarrow b \mid bS \mid aBB$ For the string "aaabbabbba" Find a a. Leftmost derivation b. Rightmost derivation c. Parse tree	CO3	BT3
Q.10	Explain the process of tokenization in detail.	CO5	BT2
Q.11	Explain different optimization techniques in detail.	CO4	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT): BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the different phases of compilation
CO 2	Describe the design of a compiler, including its phases and components, and develop a large, complex, but well-structured software system that implements various phases of a compiler, such as the scanner, parser, code generator, and optimizer.
CO 3	Apply context-free grammars to syntax analysis
CO 4	Analyze optimization techniques to reduce the code size
CO 5	Develop skills in lexical analysis and parsing, which are helpful to a broad range of computer science application areas.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Summer Term Examination 2024-25

T.Y. B. Tech.

Computer Science and Engineering

Database management system

PCC-CS302

Date:19/06/25

Time: 10:00-13:00

Max. Marks :100 Marks

Q.N.		Questions (ATTEMPT ALL)	Marks	CO	BT
Q-1	a)	Which of the following is not a type of database? a) Hierarchical b) Network c) Distributed d) Decentralized	02	CO1	BT1
	b)	The DBMS acts as an interface between _____ and _____ of an enterprise-class system. a) Data and the DBMS b) Application and SQL c) Database application and the database d) The user and the software	02	CO1	BT2
	c)	The ability to query data, as well as insert, delete, and alter tuples, is offered by _____. a) TCL (Transaction Control Language) b) DCL (Data Control Language) c) DDL (Data Definition Language) d) DML (Data Manipulation Language)	02	CO2	BT1
	d)	_____ is a set of one or more attributes taken collectively to uniquely identify a record. a) Primary Key b) Foreign key c) Super key d) Candidate key	02	CO3	BT2
	e)	What is DBMS? a) DBMS is a collection of queries b) DBMS is a high-level language c) DBMS is a programming language d) DBMS stores, modifies and retrieves data	02	CO2	BT1
Q-2		Define the following:-			
	a)	Define foreign key with example.	02	CO2	BT2
	b)	What is data definition language?	02	CO3	BT1

	c)	Define serializability.	02	CO4	BT1
	d)	What is derived attribute ?	02	CO3	BT2
	e)	What is functional dependency?	02	CO5	BT2
Q-3		Explain database management system architecture with neat diagram.	10	CO2	BT1
Q-4		Elaborate on 1NF, 2NF & 3NF with suitable examples.	10	CO4	BT2
Q-5	a)	Convert following ER diagram to Relational model.	05	CO3	BT5
		<pre> graph LR subgraph Scientist FName((FName)) LName((LName)) SName((SName)) SId((SId)) RArea((RArea)) Country((Country)) end subgraph Invention IId((IId)) IName((IName)) Year((Year)) end Scientist --- Invents{Invents} --- Invention style SId stroke-width:4px style IId stroke-width:4px </pre>			
	b)	Explain any four aggregate functions with example.	05	CO4	BT2
Q.6	a)	Define transaction. Explain briefly different states of transition with neat state transition diagram.	05	CO4	BT2
	b)	Explain ACID properties of transaction.	05	CO1	BT1
Q.7	a)	Explain any 2 locking protocols.	05	CO4	BT2
	b)	Write the SQL queries for following EMP table. Emp (empno, deptno, ename, salary, designation, city.) i) Display average salary of all employee. ii) Display names of employees who stay in Mumbai or Pune iii) Set the salary of employee 'Ramesh' to 50000. iv) Display names of employees whose salaries are less than 50000 v) Remove the Record of employees whose deptno is 10 vi) Remove the column deptno from EMP table.	05	CO2	BT5
Q.8	a)	Draw an ER diagram for library management system. (Use Books, Publisher & Member entities).	05	CO3	BT4
	b)	Explain DDL and DML commands in MySQL.	05	CO2	BT3
Q.9	a)	Explain join strategies in detail.	05	CO1	BT4
	b)	Explain access control and its models.	05	CO4	BT1

Q.10	Write SQL queries for following. Consider table stud (roll no, name, sub1, sub2, sub3).	10	CO3	BT5
	i) Display names of student who got minimum mark in sub1.			
	ii) Display names of students who got above 40 marks in sub2.			
	iii) Display count of Students failed in sub2.			
	iv) Display average marks of sub1 of all students.			
	v) Display names of students whose name start with 'A' by arranging them in ascending order of sub1 marks.			
	vi) Display student name whose name ends with 'h' and subject 2 marks are between 60 to 75.			

*****ALL THE BEST*****

COURSE OUTCOMES:

1. CO1: Understand the role and issues in management of data such as efficiency, privacy, security, ethical responsibility, and strategic advantage.
2. CO2: Create, maintain and manipulate a relational database using SQL.
3. CO3: Describe ER model and normalization for database design.
4. CO4: Examine issues in data storage and query processing and can formulate appropriate solutions.
5. CO5: Design and build database system for a given real world problem

Bloom's taxonomy(BT)

BT1- Remember BT2-Understand BT3-Apply BT4-Analyze BT5-Evaluate BT6-Create